

# Lecture 8: OSI Model — In Depth

*The Data Link Layer handles MAC addressing, ARP, framing, and flow/access control — and the Network Layer builds on top of it with four core responsibilities of its own.*

## MAC Address — In Depth

**MAC Address = Media Access Control Address**, also known as the **Physical Address**.

- ♦ It is **unique** to each device and **does not change** under normal circumstances (it can technically be changed/spoofed, but this is considered **illegal**).
- ♦ Every device — laptop, phone, printer, etc. — has a MAC address, and the number of MAC addresses a device has depends on its NIC (the number of ways it can connect to the internet).

### Format of MAC Address

48-bit number, represented in Hexadecimal form

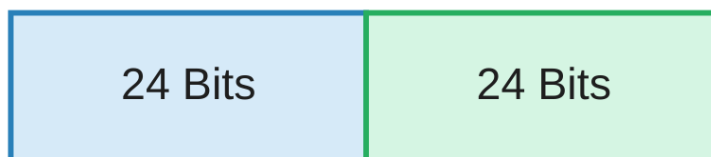
Minimum: 00:00:00:00:00:00

Maximum: FF:FF:FF:FF:FF:FF

Each hex digit = 4 bits

6 groups × 8 bits per group = 48 bits total

### How MAC Addresses Are Distributed



- ♦ The **first 24 bits** are assigned to a specific **organization** (manufacturer).
- ♦ The **remaining 24 bits** let that organization create up to  **$2^{24}$  unique devices**.

**Examples:**

| ORGANIZATION | ASSIGNED PREFIX |
|--------------|-----------------|
| Google       | A0:2C:33        |
| Cisco        | CC:46:D6        |
| Oracle       | C3:4:28:6F      |

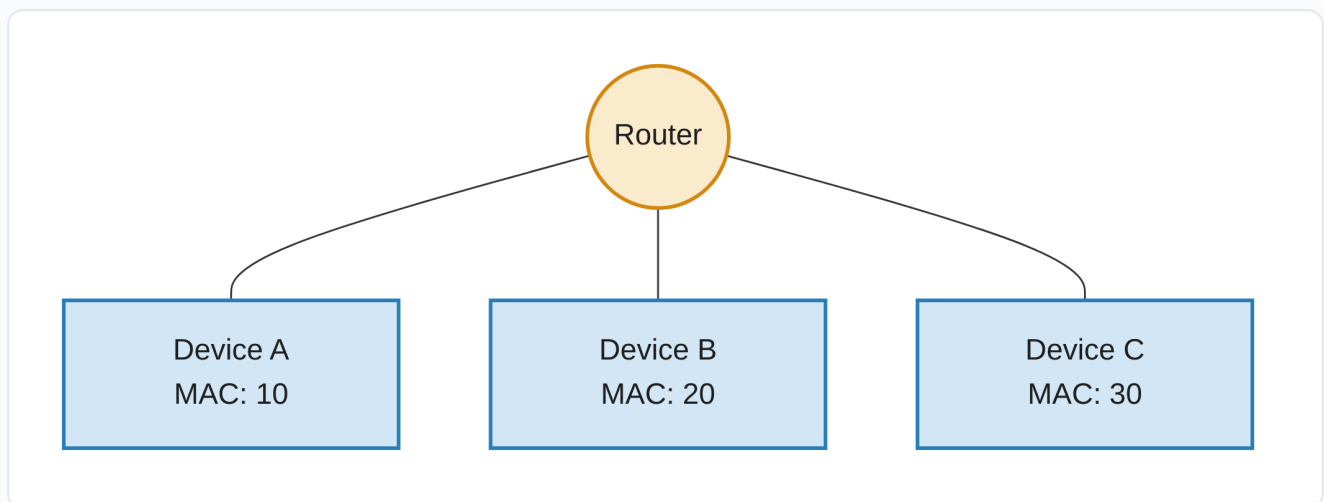
Each organization can generate  $2^{24}$  **devices** starting from its assigned prefix — a MAC address is therefore a **12-digit hexadecimal number**, which is fundamentally a **48-digit binary number**.

## Notations of MAC Address

MAC addresses can be written in three equivalent formats:

1. Hyphen-Hexadecimal: 00-0a-83-b1-c0-8e
2. Colon-Hexadecimal: 00:0a:83:b1:c0:8e
3. Period-Separated Hexadecimal: 000.a83.b1c.08e

## How Are Other Devices' MAC Addresses Known Within a LAN?

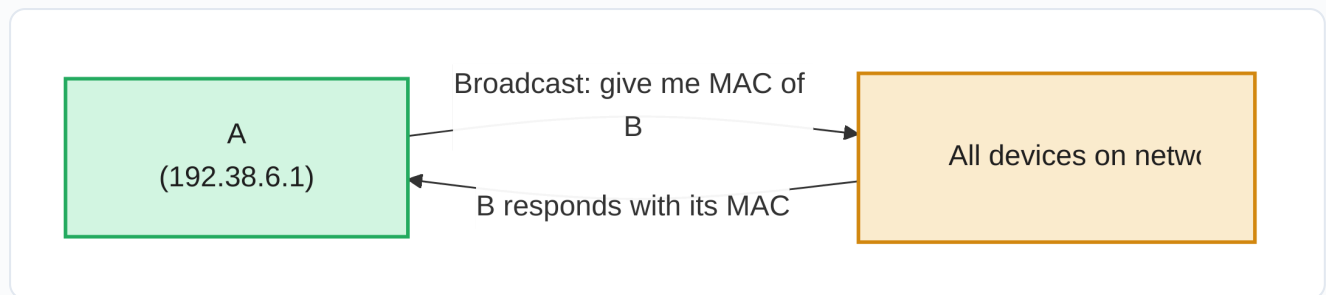


- ✦ **Device A can transfer/send data using the MAC address** of any device in that particular network.
- ✦ The **router assigns Private IP Addresses**, and those private IPs are known to the devices — but their **MAC addresses** still need to be discovered separately.

## ARP — Address Resolution Protocol

**ARP (Address Resolution Protocol)** finds the **MAC address** of a device when only its **IP address** is known.

**Scenario:** Device A wants to send data to Device B, but only knows B's **IP address**, not its MAC address.



1. **A broadcasts** a message using B's IP address, requesting: *"Whoever has this IP, please respond with your MAC address."*
2. **B recognizes** its own IP in the request and **responds with its MAC address**.
3. Now A has B's MAC address and can **send data directly**.

## Broadcasting

Whenever an IP address block is allocated to a network, it comes with a **Network Address** and a **Broadcast Address**:

23.0.0.0 → Network Address  
23.255.255.255 → Broadcast Address

## How Does an ISP Trace Illegal Activity to a Specific Device?

A device's IP address can change over time — especially with a **dynamic IP address** — so the IP address alone isn't enough to permanently identify a device. To solve this, the ISP also keeps track of each device's **MAC address**, since a device connects directly to the ISP's network hardware and its MAC address stays visible at that point.

- ✦ The ISP maintains a log of which IP address was assigned to which MAC address, at what time.
- ✦ If illegal activity is traced back to a specific IP address, the ISP can check this log to find out exactly which device (identified by its MAC address) was using that IP address at that particular time.

## Error Detection & Correction

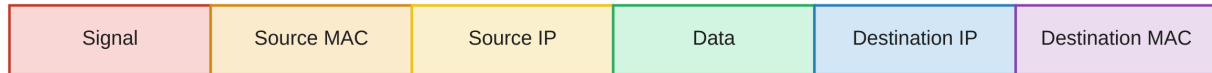
A responsibility of the Data Link Layer — detecting when a **bit error** occurs during transmission (e.g. a 0 flips to a 1), and either **correcting the error** or requesting a **resend**.

Bit sent: 0 - 0 - 0  
Bit received: 0 - 1 - 0 ← error occurred

DLL response: correct the error, or ask sender to resend the data.

## Framing

**Framing** defines the **structure** in which data actually travels across the Data Link Layer.



### How does the receiver know when data is about to arrive?

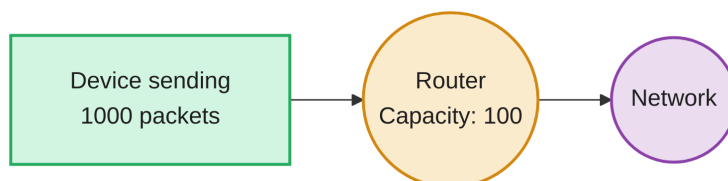
- ✦ Normally, the signal on the line is **idle** (represented as all 000000).
- ✦ When the sender is about to transmit a message, it sends a distinctive **alert signal**, e.g. 1010101011.
- ✦ As soon as the receiver detects this **pattern** (specifically, recognizing 11 at the end), it switches into an "**alert mode**" and prepares to read the **actual incoming data**.

[ A ] --- 1010101...11 ---> [ B ] (B becomes alert and ready to receive)

*This pattern-based signaling is what tells the receiver: "real data is coming right after this."*

## Flow Control

Ensures that data is sent to a device **at a rate it can actually handle**, preventing overload.

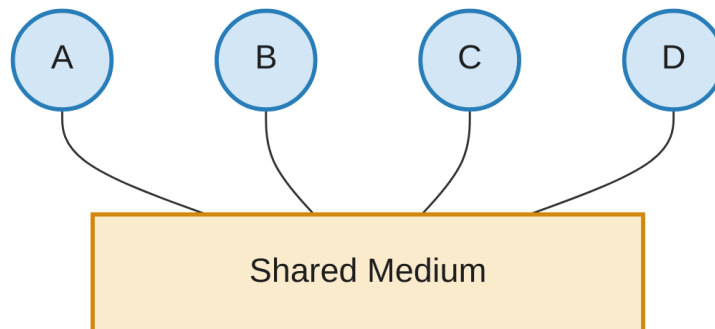


- ✦ If a router's capacity is only **100 packets**, but a device sends **1000 packets** at once, the router **cannot handle the excess**, leading to **data loss**.
- ✦ **Solution:** don't send all packets at once — send the **desired amount of data in a controlled sequence**.

**Flow Control** = sending a controlled amount of data to a device, in proper sequence, matching the receiver's capacity.

## Access Control

Manages **which device gets to transmit at any given moment** when multiple devices share the same medium.



- ♦ If multiple devices try to send signals over the same medium **simultaneously**, their signals can **collide**, potentially **corrupting the data**.
- ♦ **Solution:** allow **only one device to transfer data at a time**.

Example order: D → C, then B → A (one transmission completes before the next starts)

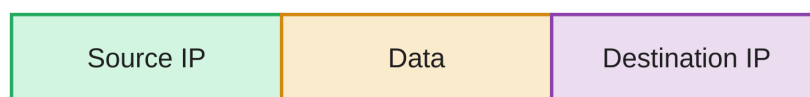
This controlling of device access is handled by the **Data Link Layer** and is called **Accessing** — it improves the overall **efficiency of the shared cables/medium**.

## 3. Network Layer

The Network Layer has **4 core responsibilities**:

### 1) Assigning Logical Address

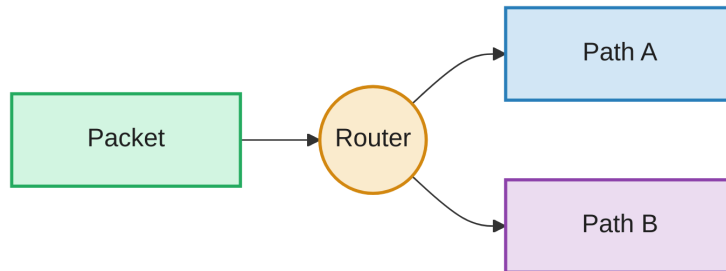
**Logical Address = IP Address** — this is the layer that gives a device its IP Address.



This is what connects the sender and receiver across the full network — unlike the Data Link Layer, which only handles a single hop at a time.

## 2) Routing

**Routing** = deciding the **route/path** a packet should take to reach its destination.



A router operates using 3 layers — Network, Data Link, and Physical — and the routing decision (which direction to send the packet) is made at the **Network Layer**.

## 3) Fragmentation & Re-Assembly

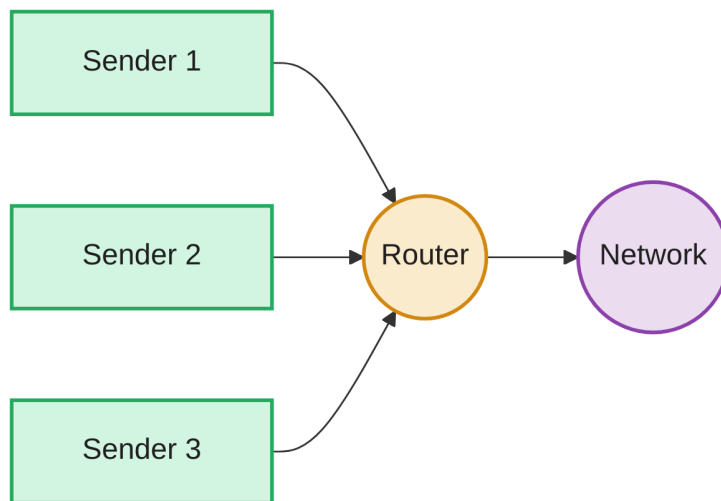
Data Link Layer unit: Frame  
Network Layer unit: Packet

**Scenario:** A packet of data (e.g. 5 MB) needs to travel across a network segment that can only handle a smaller capacity (e.g. 2.5 MB at a time).



- ♦ The original packet is **divided into smaller fragments** to fit within the network's capacity.
- ♦ **After reaching the destination, these fragments are re-assembled** back into the original data.

## 4) Congestion Control



- ♦ If **all senders transmit data at once**, excessive **traffic** is created, leading to:
  - ♦ **Delay**
  - ♦ **Collision**
  - ♦ **Data loss**
- ♦ **Solution:** there needs to be a **systematic way (protocols)** that governs how and when data is sent, to avoid overwhelming the network.

### Key Terms:

| TERM       | MEANING                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------|
| Bandwidth  | The <b>maximum speed</b> at which data can theoretically be transmitted over a network                                         |
| Throughput | The <b>actual speed</b> at which data is currently being transferred (often lower than bandwidth due to real-world conditions) |

## Summary

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- ♦ A MAC Address (Media Access Control Address / Physical Address) is a unique, generally unchanging 48-bit hexadecimal identifier assigned to each device's network interface, with the first 24 bits identifying the manufacturer and the remaining 24 bits identifying the specific device.
- ♦ MAC addresses can be written in hyphen-hexadecimal, colon-hexadecimal, or period-separated hexadecimal notation — all representing the same underlying value.
- ♦ ARP (Address Resolution Protocol) resolves a device's MAC address from its known IP address by broadcasting a request across the local network, to which the target device responds with its MAC address.
- ♦ The Data Link Layer also handles Error Detection & Correction (catching and fixing/resending corrupted bits), Framing (structuring data with a recognizable alert signal so the receiver knows when real data is arriving), Flow Control (sending data at a rate the receiver can handle), and Access Control (ensuring only one device transmits over a shared medium at a time to avoid collisions).
- ♦ The Network Layer has four core jobs: assigning the Logical (IP) Address to connect sender and receiver, Routing (deciding a packet's path, done at the router), Fragmentation & Re-Assembly (splitting large packets to fit network capacity and reconstructing them at the destination), and Congestion Control (managing traffic to avoid delay, collisions, and data loss).
- ♦ Bandwidth is the maximum possible data transmission speed over a network, while Throughput is the actual speed being achieved in practice.